INTERHEMISPHERIC EEG ASYMMETRY AS A NOVEL BIOMARKER FRAMEWORK FOR SEIZURE PREDICTION USING RANDOM FOREST CLASSIFICATION

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Abstract:

Epilepsy affects ~1% of people worldwide, and seizure unpredictability severely impacts quality of life. Forecasting aims to detect the pre-ictal stage, but subtle variability makes it hard to distinguish from inter-ictal activity. Traditional EEG features capture only part of this complexity, with above-chance prediction possible in only ~10% of patients. In this study, we investigated whether interhemispheric EEG asymmetries could serve as novel biomarkers for seizure prediction.

Using the CHB-MIT dataset (12 patients, 78 seizures, ~344 hours), we derived asymmetry indices (R-L/R+L) from bipolar channel pairs and evaluated them against baseline models with a Leave-One-Patient-Out cross validation (LOPO-CV) based Random Forest classification. Performance was measured at the event level (sensitivity, FPR/hr) and window level (sensitivity, specificity, F1). Statistical tests included paired t-tests, repeated-measures ANOVA, and post-hoc Dunnett analyses.

Although 14 significant asymmetries were identified across five bipolar channels, only three— all within the FP1–F7 region (FP-FM)—were found to significantly influence model performance. Notably, all 14 asymmetries exhibited a rightward shift, highlighting the right hemisphere's potential role in seizure buildup Removing all asymmetry features reduced both sensitivity and FPR/hr, while ablating only these three features preserved sensitivity but markedly increased FPR/hr. This indicates that specific asymmetries mainly reduce false positives, while asymmetry features overall enhance predictive performance. The combined model achieved strong results (sensitivity ~0.85, FPR/hr ~0.27), indicating that interhemispheric asymmetries provide robust complementary markers for seizure forecasting. These findings highlight the potential of asymmetry-based features to advance clinically viable prediction systems.

Background Information:

- Prior studies have extracted time, frequency, and time-frequency features to classify seizure vs non-seizure segments of EEG data
- However, interhemispheric asymmetries are largely underexplored for their role in predicting seizures
- While not seizures themselves, IEDs exhibit hemispheric asymmetries, suggesting that interhemispheric differences in EEG activity may serve as informative features for seizure prediction (Loddenkemper et.al, 2007)

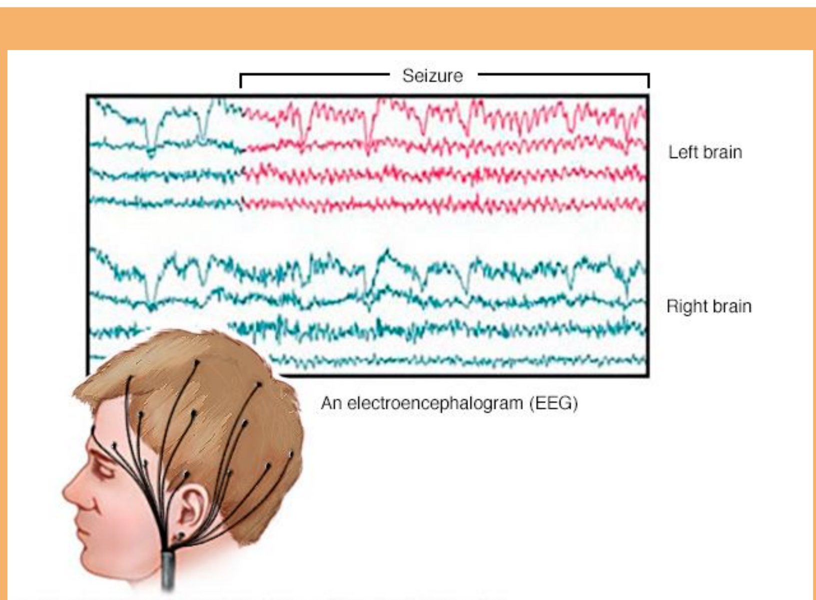

Figure 1: EEG traces illustrating typical seizure activity with annotations showing pre-ictal and ictal stages of a seizure

Method/Materials:

- Database
  - CHB-MIT EEG Database
- Filtering
  - 5th Order Low Pass Butterworth Filter (0.5 - 45 Hz)
- Feature Extraction
  - Time-Domain Features:, Hjorth Parameters (Activity, Mobility, Complexity)
  - Frequency-Domain Features
    - Bandpower (Delta, Theta, Alpha, Beta, Gamma)
    - Computed via Welch's Method
  - Asymmetry Index
    - Measures relative differences between left and right hemisphere EEG signals.
    - Computed as (R-L)/(R+L) for homologous bipolar channel pair
  - 10-minute pre-ictal period, 30-second time windows
- Random Forest Classifier
  - Patient-specific Leave-One-Patient-Out Cross-Validation (LOPO-CV)
  - Model metrics (sensitivity, specificity, F1-score, FPR/hr)

Results:

Model Performance Metrics:

- Sensitivity, specificity, F1-score, and FPR/hr were calculated to assess model performance when asymmetries were removed
- Performance was compared for models with selective removal of asymmetries and with the removal of all asymmetries.

Hemispheric Shifts:

- Features exclusively shifted towards the right hemisphere during the buildup to a seizure.
- The most significant asymmetries were observed in the FP1-F7 and FP2-F8 regions, with a clear trend toward right hemisphere dominance.

Impact of Asymmetries on False Positives:

- The most significant asymmetries exclusively affected the detection of false positives.
- Removing these asymmetries led to a decrease in FPR/hr without compromising other metrics.

Broad Analysis of Asymmetries:

- When asymmetries across all features were considered, there was a notable improvement in all performance metrics (sensitivity, specificity, F1-score, and FPR/hr).

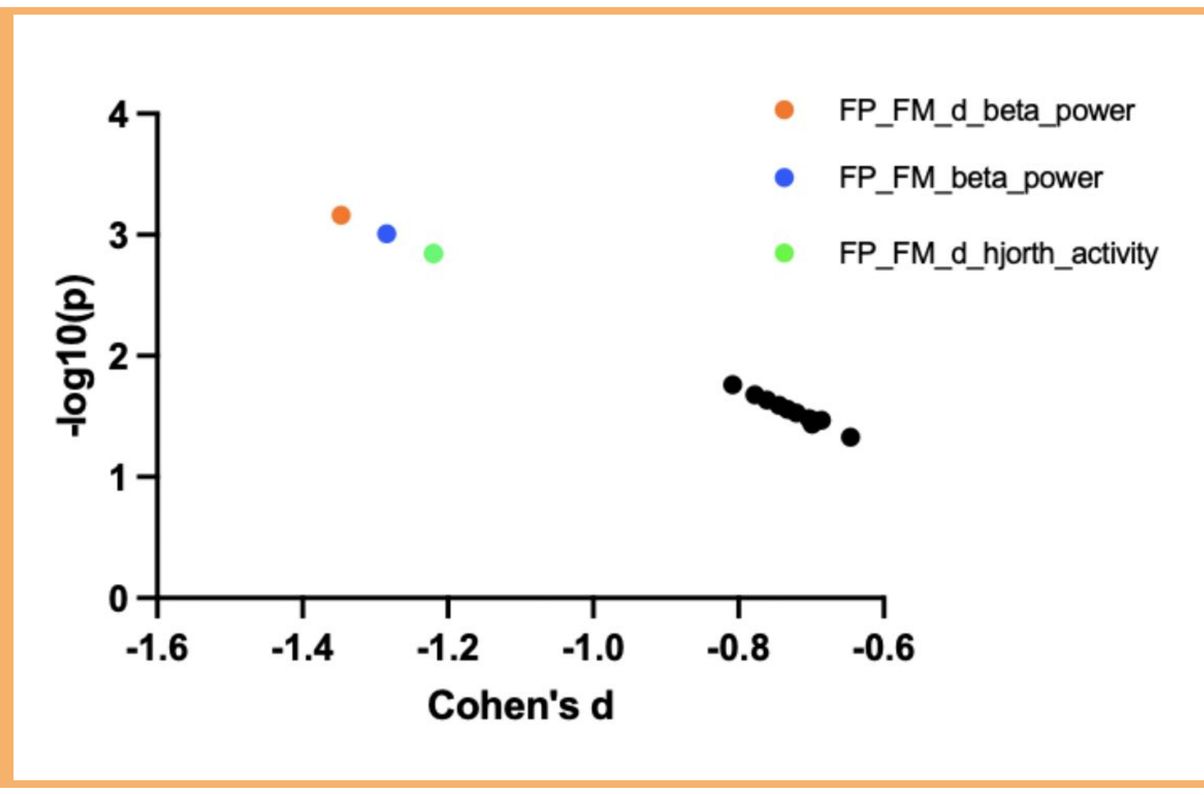

Figure 2. Plot of -log$_{10}$(p) vs Cohen's d for most significant asymmetries
Beta power, the derivative of beta power, and the derivative of Hjorth Activity were more prominent compared to other asymmetries with statistical significance.

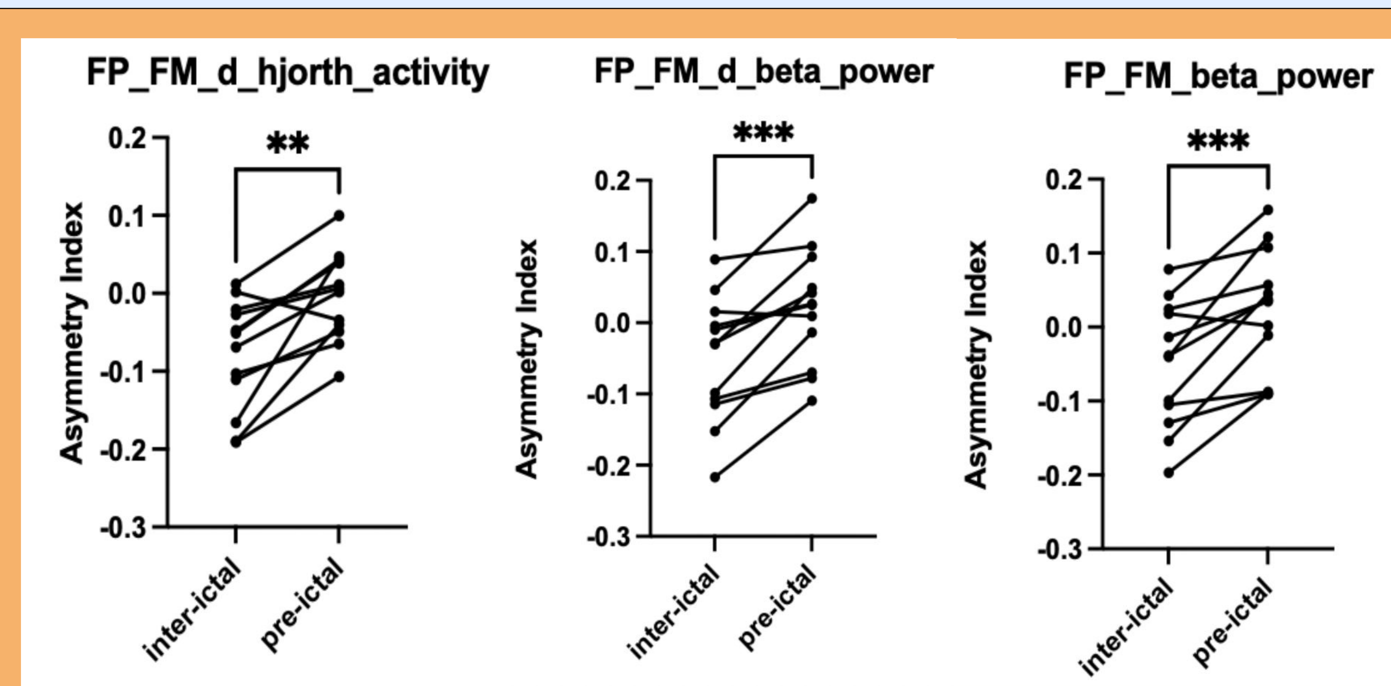

Figure 2: **Top three asymmetrical changes resulted in shifts towards right hemisphere**
Statistical analysis from paired t-test shows significant increase in asymmetry index. (*p < 0.05, **p < 0.01, and ***p < 0.001). These results remained significant after False Discovery Rate (FDR) correction.

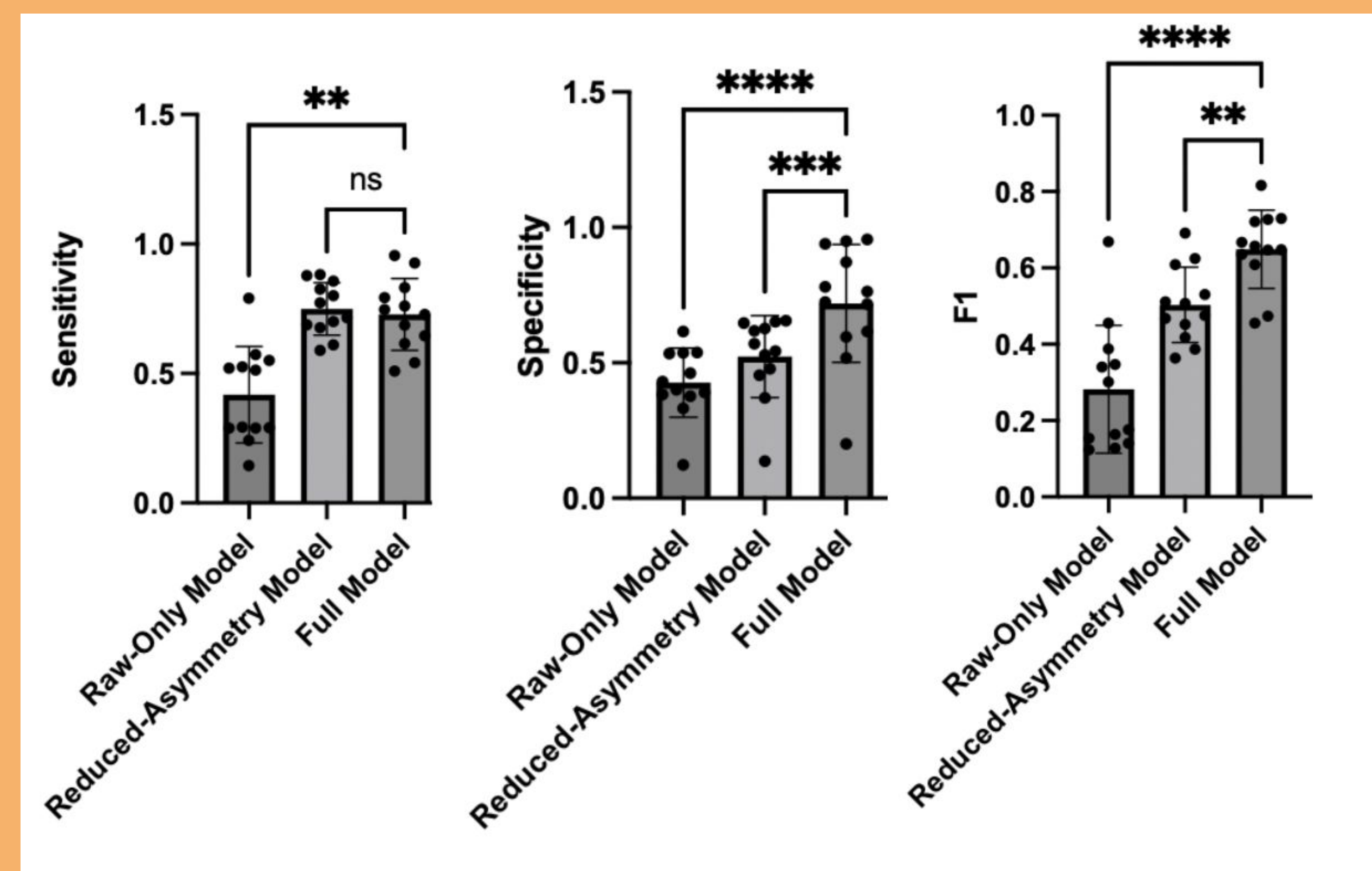

Figure 3: Asymmetries significantly impact model performance, but partial ablation only alters specific metrics in window-level analysis.
Ablating top three asymmetries reduces specificity and F1-score, but maintains sensitivity. Excluding all asymmetries significantly improves overall model performance. Results remained significant after FDR correction following post-hoc Dunnett's test in a one-way ANOVA

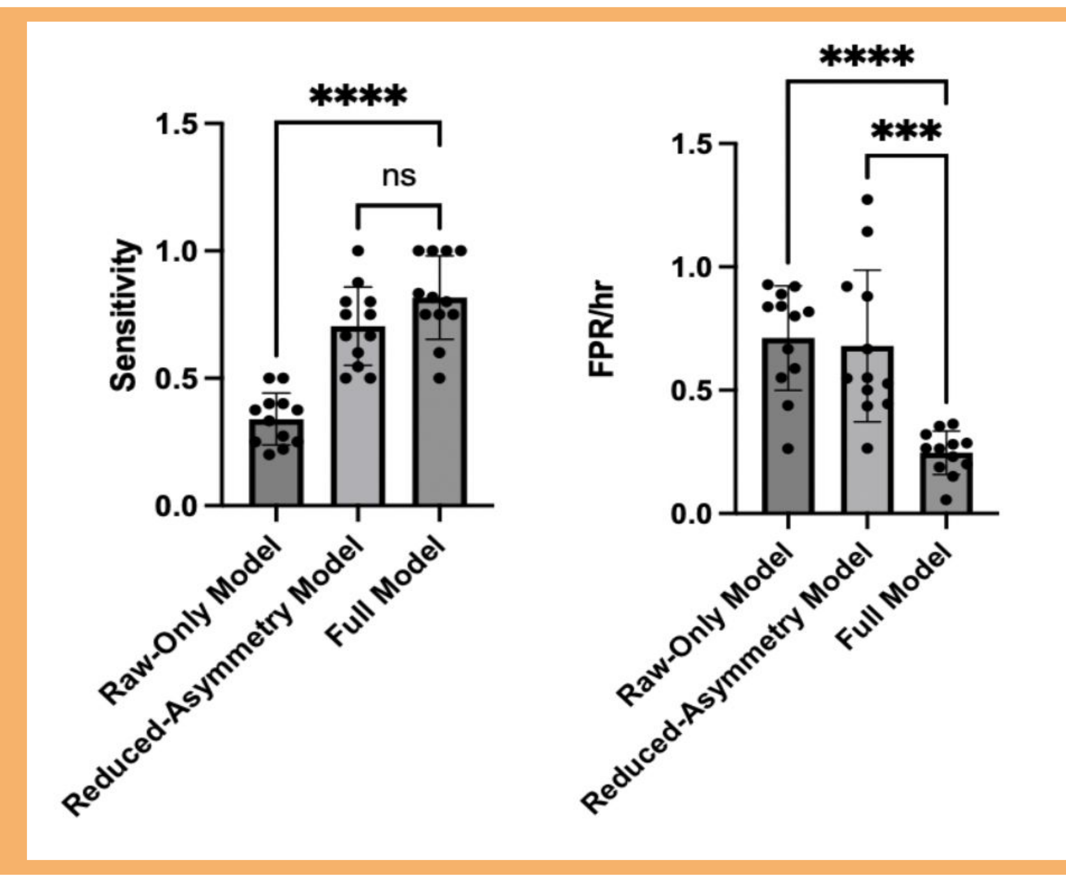

Figure 4: **Asymmetries similarly improve sensitivity and false positives at event-level detection while partial ablation exclusively improves false positives.** Results remain significant after FDR correction following post-hoc Dunnett's test in a one-way ANOVA.

Discussion/Conclusion:

- **Cohen's d for significant changes in asymmetry index from inter-ictal to pre-ictal were all negative**
  - -> Asymmetry tends to shift towards right hemispheres
- **Top 3 asymmetries were all found in the FP1.F7 & FP2.F8 pairs**
  - -> Fronto-polar and fronto-medial regions are implicated in significant interhemispheric differences in the buildup to a seizure
- **In the window-level analysis, specificity and F1-score were significantly worse when ablating the top three asymmetrical features or all asymmetries**
  - -> Specific asymmetries and asymmetries in general can both be incorporated into models to reduce false positives and improve overall performance
- **Sensitivity only decreased when removing asymmetries entirely**
  - -> While asymmetries as a whole drastically improve ability to predict seizures, specific features are insufficient on their own

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